

AKSHAYA THOUTAM

akshaya_thoutam@hms.harvard.edu | akshayathoutam.github.io | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

EDUCATION

HARVARD MEDICAL SCHOOL M.S Clinical Research

Cambridge, Ma 2025-Present

- Relevant Coursework: CI 700: Ethics and the IRB, CI 701: Clinical Data Science Design and Analytics I, CRES 701: Organized Basic Evidence-Based Medicine, CI 702A: Clinical Trials (Part 1), CRES 703: Developing Therapies for Patients, CRES 705: Systematic Review and Meta-Analyses, CRES 709: Interpreting Medical Literature (Part 1)

GEORGIA INSTITUTE OF TECHNOLOGY B.S Neuroscience

Atlanta, Ga 2020-2023

- Minor in Health and Medical Sciences
- Relevant Coursework: Linear Algebra, Integral Calculus, Biostatistics, Machine Learning in Bioscience, Cancer Biology and Technology, Tumor Microenvironment and Immuno-Oncology (MIT), Advanced Computational Biology: Genomes Networks Evolution (MIT), Statistical Inference for Brain and Cognitive Sciences (MIT), Artificial Intelligence in Medicine II (Harvard)

PUBLICATIONS

1. Zeinab Navidi, **Akshaya Thoutam**, Madeline Hughes, Srivatsan Raghavan, Peter Winter, Lorin Crawford, and Ava P. Amini. Adaptive Resampling For Improved Cellular Representation Learning in Imbalanced Single-cell Samples. (2025). *Peer review – Nature Communications*.
2. Alan DenAdel, Madeline Hughes, **Akshaya Thoutam**, Ava P. Amini, and Lorin Crawford. Evaluating the role of pre-training dataset size and diversity on single-cell foundation model performance. (2024). *Peer Review – Nature Methods*.
3. Ajay Nadig & **Akshaya Thoutam**, Madeline Hughes, Srivatsan Raghavan, Peter Winter, Ava P. Amini, and Lorin Crawford. Consequences of training data composition for deep generative models in single cell biology (2025). *Peer Review – Genome Biology*.
4. Greg Gibson, Dermot McGovern, **Akshaya Thoutam**, the NIDDK IBD Genetics Consortium, Judy Cho and John Rioux. Eleven Grand Challenges for Inflammatory Bowel Disease Genetics and Genomics. (2024). *Inflammatory Bowel Disease*
5. Nasrin Hooshmand, **Akshaya Thoutam**, Max Anikovskiy, Hagar Labouta C Mostafa El-Sayed (2021). Localized Surface Plasmon Resonance as a Tool to Study Protein Corona Formation on Nanoparticles. *Journal of Physical Chemistry C*, 125(45), 24765–24776
6. **Akshaya Thoutam**, Narasiah C Kolliputi (2021). Epigenetics of pulmonary diseases. In T. Tollefsbol (Ed.), *Medical Epigenetics* (2nd ed., pp. 45-67). Publisher. ISBN: 9780128239285
7. **Akshaya Thoutam**, Mason Breitzig, Richard Lockey C Narasiah Kolliputi (2020). Coronavirus: a shift in focus away from IFN response and towards other inflammatory targets. *Journal of cell communication and signaling*, 1–2

RESEARCH EXPERIENCE

Project Ex Vivo | **BROAD INSTITUTE OF HARVARD/MIT** **Computational Associate II**

(July 2025 – Present)

- Using multimodal machine learning with scRNA-seq and spatial transcriptomics data to define and engineer cell states in cancer

Biomedical Machine Learning – Project Ex Vivo | **MICROSOFT RESEARCH** **Predoctoral Research Assistant**

(Feb 2024 – July 2025)

- Manipulating single cell foundation models to increase data depth and biology in capturing cell-state level biology
- Creating an end-to-end pipeline for alignment and analysis of single-cell RNA seq data

Levin Lab (Genomics) | **BROAD INSTITUTE OF HARVARD/MIT** **Computational Associate Intern**

(May 2023 – September 2023)

- Integrated long-read MAS-seq data, Visium spatial transcriptomics data, and single-nucleus RNA-seq to investigate spatially organized RNA isoform diversity in a pilot study on Parkinson's disease.
- Analyzed single-nucleus RNA-seq data of a SCN2A knock-out mice cortex from the SCHEMA Schizophrenia initiative.

Knott Lab (Genomics + ML) | **CEDARS SINAI** **Computational Associate Intern**

(May 2022 – October 2022)

- Analyzed temporal and spatial differences of cell types with scRNA-seq and ATAC-seq data from patients of an ER+ Breast Cancer clinical trial to characterize the tumor microenvironment in three different time periods to elucidate the molecular mechanisms that drive ER+ breast cancer.

Gibson Lab | **GEORGIA TECH** **Computational Research Assistant**

(May 2021 – December 2023)

- Curated 133,488 B cells characterized from 246 tumor samples across 9 human cancer types to identify distinct molecular characteristics in the tumor microenvironment across cancer types - creation of pan-cancer atlas of B cells.
- Compared the Immune transcriptional spectrum across healthy colon, IBD, Crohn's Disease, and colon cancer progression.

Laser Dynamics Lab | **GEORGIA TECH** **Research Assistant**

(May 2021 – May 2022)

- Computationally modeled and synthesized gold and silver plasmonic nanoparticles of different morphologies for their applications in bio-sensing and photothermal therapy in cancer treatment

Kolliputi Lab (Immunology) | **UNIVERSITY OF SOUTH FLORIDA** **Research Writer**

(May 2020 – March 2021)

- Compared respiratory viruses MERS-CoV, SARS-CoV-1, HPIV3, RSV, IAV to SARS-CoV-2 to uncover monocyte cytokine production as a driving feature of COVID-19 infection.
- Invited to write the chapter on 'The Epigenetics of Pulmonary Disease' for the textbook Medical Epigenetics, 2E (Elsevier)

POSTERS

1. **Akshaya Thoutam**, Madeline Hughes (2025, April). Evaluating the role of pre-training dataset size, diversity, and composition on single-cell foundation model performance. [RECOMB 2025 Research in Computational Molecular Biology](#).
2. **Akshaya Thoutam**, William Watson, Mohan William, Regina Nwosu (April 2023). 5HT2c against lorcazerin on spontaneous nerve root 3 activity in crayfish model. [Georgia Tech Undergraduate Neuroscience Symposium](#)
3. Hira Anis, **Akshaya Thoutam**, Makeda Hailu, Vishal Dhere, Greg Gibson, Erin Connolly (2023, April) Single-cell Pan-cancer atlas reveals diversity and plasticity of tumor-infiltrating plasma cells. [Atlanta Workshop for Single Cell Omics 2023](#)
4. **Akshaya Thoutam** C Greg Gibson (2022, October). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating B cells. [National Symposium for Undergraduate Research](#)
5. **Akshaya Thoutam**, Erin Connolly, Greg Gibson (2022, April). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating Plasma Cells. [Atlanta Workshop for Single Cell Omics 2022](#).
6. **Akshaya Thoutam**, Erin Connolly, Greg Gibson (2022, April). A Pan-cancer Single-cell Transcriptional Atlas of Tumor-infiltrating Plasma Cells. [Atlanta Workshop for Single Cell Omics 2022](#).

TALK

1. **Akshaya Thoutam** & Greg Gibson (2023, April). Pan-cancer atlas of tumor-infiltrating B cells. Georgia Institute of Technology [Undergraduate Research Symposium](#)

AWARDS/HONORS

BroadHacks 2025 Broadie's Choice Award Winners – Bronze Broad Institute	June 2025
College of Sciences Deans Scholar Georgia Institute of Technology	Awarded every semester
Early Research Award Georgia Institute of Technology	August 2021
Robert Gunn Student Award (Finalist) American Physiology Society	April 2023

LEADERSHIP

Down to Dance (Founder/President) Georgia Institute of Technology	August 2020 – May 2023 Urban
Health and Wellness committee leader GT AMSA	September 2020 – May 2021
Fundraising committee leader GT AMSA	September 2020 – May 2021

SHADOWING

Pediatric Pathology | Boston Children’s Hospital | **60 hours**
Pediatric Heme Oncology | Boston Children’s Hospital | **8 hours**
Neurosurgery | Atlanta Veteran Affairs Medical center | **80 hours**
Pediatrician | All Pediatrics Tampa | **100 hours**
Sarcoma Oncology | Moffitt Cancer Center | **40 hours** **Thoracic surgery** | Moffitt Cancer Center | **40 hours**

VOLUNTEERING

Emergency Department Massachusetts General Hospital 74 hours	August 2024 - Present
Arts in medicine Program Emory Winship Cancer Center 150 hours	August 2020 – May 2021
Care package Advent Health Florida 100 hours	March 2020 – December 2020
Recycling initiative Cedars Sinai Hospital 200 hours	May 2022 – August 2022